Reproducibility Artefact – Functional Evaluation
Synthetic Patient History and User Story Evaluation

Purpose and scope

This supplementary document describes the synthetic longitudinal patient history used for the functional evaluation and illustrates the expected observable outcome for each of the eight clinician-centred user stories. The document complements the machine-readable synthetic dataset and the functional traceability artefacts in the accompanying reproducibility package.

User stories

The following user stories articulate clinician-facing functional needs, independent of any specific openEHR modelling approach. They frame the functional evaluation by specifying the observable outcomes and navigation patterns expected of an episode-oriented record. In this study, “health problem” (diagnosis/problem) is treated as synonymous with an episode of care; accordingly, the stories assume the ability to pivot between chronological, section-based, and episode-centred views. Each story is operationalised using the same synthetic patient history to demonstrate the expected behaviour.

1. As a healthcare professional, I would like to see all medical record entries for a patient in the CDR displayed chronologically and grouped by contact.
2. As a healthcare professional, I would like to be able to select an Episode of care (health problem = diagnosis/problem) and then view all medical record entries relating to that health problem in the patient's complete CDR.
3. As a healthcare professional, I would like to see all medical record entries in a patient's CDR organised according to the standard clinical sections of a medical record, for example medical history, physical examination, diagnostics, therapy, risk factors, progress notes.
4. As a healthcare professional, I would like the grouping to be according to the standard clinical sections of a medical record, with the chronological grouping by contacts or with the corresponding record entries for the selected health problem displayed.
5. As a healthcare professional, I would like to have a linear list of all patient's recorded health problems (= diagnosis/problem = episode-of-care).
6. As a healthcare professional, I would like to be able to create a hierarchical 'diagnosis and problem list' for each patient, organised in a tree structure according to Weed's problem list, which I could then adapt continuously.
7. As a healthcare professional, I would like to be able to create progress notes according to the SOAP scheme for each health problem (= diagnosis/problem = episode-of-care) or for several health problems at the same time.
8. As a healthcare professional, I would like a dashboard for each patient containing various widgets. Each widget would display medical record entries for a single section, in accordance with the standard clinical sections of a medical record.

Synthetic patient history

A synthetic longitudinal patient history was constructed specifically for the functional evaluation. The dataset comprises 121 medical record entries and includes acute episodes (e.g. urinary tract infection and cholecystitis), chronic conditions (e.g. coronary heart disease and diabetes), historical diagnoses, routine encounters, diagnostic findings, medications, and longitudinal changes in clinical status. The dataset was purpose-built to provide sufficient clinical and temporal complexity to exercise the eight predefined user stories. The same underlying dataset is used across the investigated implementation patterns to support a consistent comparison. It serves as a controlled evaluation dataset rather than as evidence of clinical effectiveness or real-world system performance.

The following case is entirely synthetic and was created solely for methodological evaluation.

Clinical introduction

Ms Anita Zbinden, 79-year-old woman (DOB 25 Dec 1944), new to the medical centre following relocation one month ago. General condition good, alert and fully orientated; pulse 75 bpm, regular.

Diagnosis and problem list

1. Coronary heart disease with:
   - Arterial hypertension (Dx 2009)
   - Heart failure
   - History of Myocardial infarction (2015)
2. Diabetes mellitus type 2 (Dx 2007) with:
   - Polyneuropathy (Dx 2014)
   - Nephropathy (Dx 2017)
   - HbA1c 23.02.2024: 6.4%
3. Obesity WHO grade II
   - Baseline BMI 35.9 kg/m2
   - Start therapy with liraglutide 03.04.2022
   - BMI 16 May 2024: 31.3. kg/m2
4. Husband in need of care
5. Hammer toe, left foot
6. Appendectomy (1965)

Current medication

- Acetylsalicylic acid (Aspirin Cardio) 100 mg tablet (1-0-0-0)
- Torasemide (Torasemid Spirig HC) 5 mg tablet (1-0-0-0)
- Valsartan (Valsartan Mepha) 160 mg tablet (1-0-0-0)
- Rosuvastatin (Rosuvastatin Sandoz) 20 mg tablet (1-0-0-0)
- Metformin (Metformin Sandoz) 1,000 mg tablet (1-1-1-0)
- Zolpidem (Zolpidem Mepha) 10 mg tablet (0-0-0-1) when required (PRN)

Current care

Comprehensive longitudinal management of chronic conditions within the practice; acute episodes – urinary tract infection and cholecystitis – are managed episodically within the episode-oriented record.

Extract from spreadsheet

The excerpt shown is taken from the spreadsheet-based synthetic patient history. The complete machine-readable dataset is provided in the accompanying reproducibility package as `03-synthetic-dataset/synthetic-patient-medical-record.xlsx` and `03-synthetic-dataset/synthetic-patient-medical-record.csv`. Each medical record entry captures narrative detail and the contextual attributes required for the evaluation, including clinical section, linked episode of care, condition type (problem/diagnosis), clinical status, and event and recording timestamps. The dataset underpins the retrieval examples and functional evaluation and enables entries to be related consistently to contacts, episodes, and clinical sections.

Date	Contact	Detail entry	Section	Health problem Episode of Care	ID	Condition	Clinical Status	Recording Timestamp	Event Timestamp	Availability Timestamp
15.11.2024	Consultation	Reason for Encounter: Initial consultation for dysuria Chief Complaint: Burning sensation with urination.	Present illness					14.11.24 16:30	15.11.24 11:00	
15.11.2024	Consultation		Problem/Diagnosis	Dysuria	10	Problem	Active	15.11.24 10:00	15.11.24 10:00	15.11.24 10:00
15.11.2024	Consultation	Blood pressure 140/90	Vital signs	General health problem	10	Problem	Active	15.11.24 10:15	15.11.24 10:10	15.11.24 11:00
15.11.2024	Consultation	Heart rate 75	Vital signs	General health problem	10	Problem	Active	15.11.24 10:15	15.11.24 10:10	15.11.24 11:00
15.11.2024	Consultation	Weight = 86 kg	Vital signs	General health problem	10	Problem	Active	15.11.24 10:15	15.11.24 10:10	15.11.24 11:00
15.11.2024	Consultation	Height = 168 cm	Vital signs	General health problem	10	Problem	Active	15.11.24 10:15	15.11.24 10:10	15.11.24 11:00
15.11.2024	Consultation	BMI = 30.5	Vital signs	General health problem	10	Problem	Active	15.11.24 10:15	15.11.24 10:10	15.11.24 11:00
15.11.2024	Consultation	Urinalysis – dipstick: Leukocyte esterase: 3+, Nitrite: positive, Blood: 1+, Protein: trace, pH: 6.0	Laboratory results	Dysuria	10	Problem	Active	15.11.24 10:30	15.11.24 10:30	15.11.24 11:00
15.11.2024	Consultation	S: Burning sensation during urination for 3 days. First episode. Denies visible blood in urine O:pyuria A: Suspected cystitis P: antibiogram, Started treatment with trimethoprim– sulfamethoxazole. Therapy adjusted per antibiogram	Progess notes	Dysuria	10	Problem	Active	15.11.24 11:10	15.11.24 11:10	15.11.24 11:10
15.11.2024	Consultation		Problem/Diagnosis	Urinary tract infection	10	Diagnosis	Active	15.11.24 11:11	15.11.24 11:11	15.11.24 11:10
15.11.2024	Consultation	Trimethoprim–sulfamethoxazole (TMP 160 mg / SMX 800 mg) PO 2x1	Medication	Urinary tract infection	10	Diagnosis	Active	15.11.24 11:12	15.11.24 11:12	15.11.24 11:12
15.11.2024	Consultation	S: Bilateral, pressure-like headaches for ~2 years, usually relieved by one paracetamol 500 mg. Over the past few weeks pain intensity has increased and tablets give only partial relief. O: Neurologic exam: alert, cranial nerves II–XII intact, motor 5/5, normal gait, no sensory deficits, Romberg negative. Fundoscopy: optic discs sharp, no papillo-œdema. Head & neck: no scalp or temporal-artery tenderness A: Chronic tension-type headache with recent exacerbation P: Start headache diary	Problem/Diagnosis	Headache	200	Problem	Active	15.11.24 11:15	15.11.24 11:15	15.11.24 11:15

Figure 1 – Extract from the spreadsheet-based sample history, consultation on 15.11.2024

Functional evaluation traceability

The table below summarises how each user story is operationalised in the synthetic dataset and identifies the observable view used to demonstrate the expected behaviour. It provides a human-readable complement to `04-functional-evaluation/functional-requirements.md` and `04-functional-evaluation/functional-traceability.csv` in the reproducibility package.

User story	Evaluation objective	Required relationship / view	Illustrated in
US1	Chronological navigation	Entries → contact → time	Figure 2
US2	Episode reconstruction	Entries → selected episode across contacts	Figure 3
US3	Clinical organisation	Entries → clinical section	Figure 4
US4	Combined navigation	Contact or episode × clinical section	Figure 5
US5	Episode overview	Derived linear episode list	Figure 6
US6	Problem-oriented organisation	Hierarchical episode/problem relationships	Figure 7
US7	Problem-oriented progress documentation	SOAP note → one or multiple episodes	Figure 8
US8	Patient overview	Section-specific derived views	Figure 9

User stories — illustrated examples

This subsection repeats each user story and, directly beneath it, presents a figure derived from the synthetic patient history that illustrates the expected output. The figures are illustrative and implementation-agnostic; they convey the required data slices and groupings (e.g. chronological by contact, episode pivots, section-based views, and derived lists) rather than prescribing a user interface. All illustrated views are derived from the same underlying clinical record entries and their contextual relationships; no duplicate clinical source data are introduced for the purpose of generating these views.

User Story 1 – Chronological by contact

As a healthcare professional, I would like to see all medical record entries for a patient in the CDR displayed chronologically and grouped by contact.

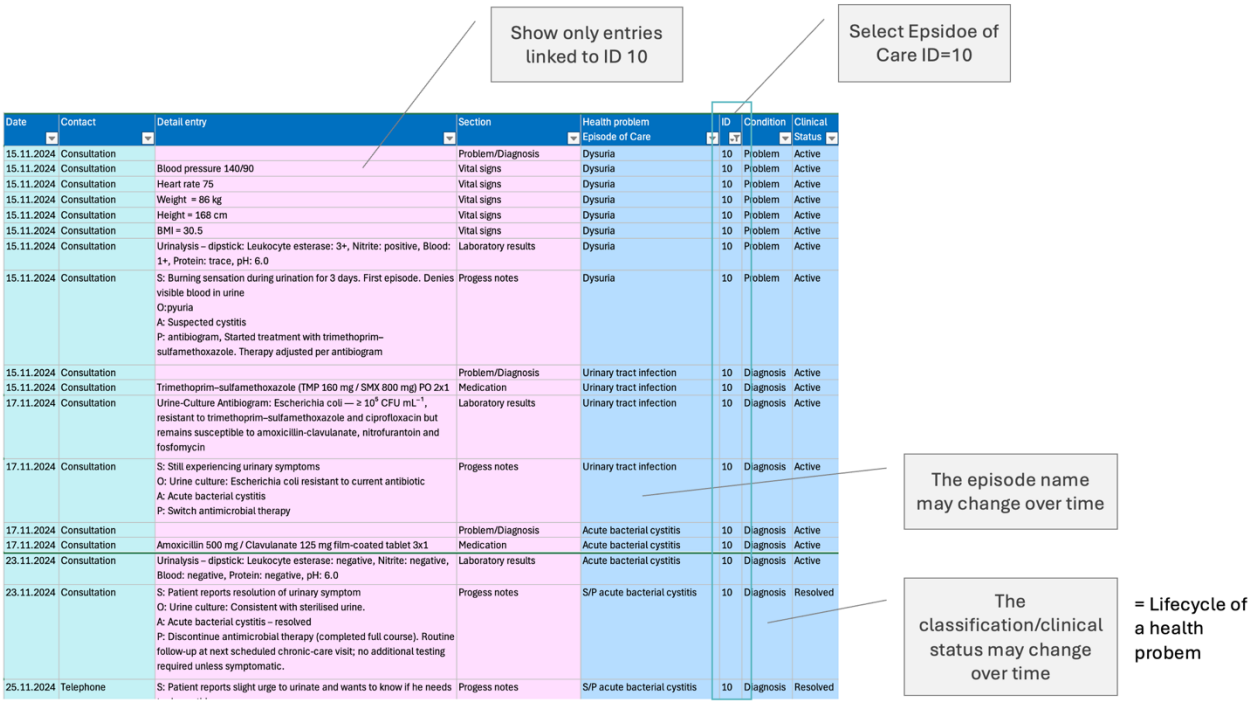


Date	Contact	Detail entry	Section	Health problem Episode of Care	ID	Condition	Clinical Status
15.11.2024	Consultation	Reason for Encounter: Initial consultation for dysuria Chief Complaint: Burning sensation with urination.	Present illness				
			Problem/Diagnosis	Dysuria	10	Problem	Active
		Blood pressure 140/90	Vital signs	Dysuria	10	Problem	Active
		Heart rate 75	Vital signs	Dysuria	10	Problem	Active
		Weight = 86 kg	Vital signs	Dysuria	10	Problem	Active
		Height = 168 cm	Vital signs	Dysuria	10	Problem	Active
		BMI = 30.5	Vital signs	Dysuria	10	Problem	Active
		Urinalysis – dipstick: Leukocyte esterase: 3+, Nitrite: positive, Blood: 1+, Protein: trace, pH: 6.0	Laboratory results	Dysuria	10	Problem	Active
		S: Burning sensation during urination for 3 days. First episode. Denies visible blood in urine O:pyuria A: Suspected cystitis P: antibiogram, Started treatment with trimethoprim–sulfamethoxazole. Therapy adjusted per antibiogram	Progess notes	Dysuria	10	Problem	Active
			Problem/Diagnosis	Urinary tract infection	10	Diagnosis	Active
		Trimethoprim–sulfamethoxazole (TMP 160 mg / SMX 800 mg) PO 2x1	Medication	Urinary tract infection	10	Diagnosis	Active
		S: Bilateral, pressure-like headaches for ~2 years, usually relieved by one paracetamol 500 mg. Over the past few weeks pain intensity has increased and tablets give only partial relief. O : Neurologic exam: alert, cranial nerves II-XII intact, motor 5/5, normal gait, no sensory deficits, Romberg negative. Fundoscopy: optic discs sharp, no papillo-oedema. Head & neck: no scalp or temporal-artery tenderness A: Chronic tension-type headache with recent exacerbation P: Start headache diary	Problem/Diagnosis	Headache	200	Problem	Active
17.11.2024	Consultation	Reason for encounter: follow-up visit and chart completion					
		Urine-Culture Antibiogram: Escherichia coli — ≥ 10^5 CFU mL^{-1}, resistant to trimethoprim–sulfamethoxazole and ciprofloxacin but remains susceptible to amoxicillin-clavulanate, nitrofurantoin and fosfomycin	Laboratory results	Urinary tract infection	10	Diagnosis	Active
		Blood pressure 142/85	Vital signs	General health problem	0	Problem	Active
		Heart rate 72	Vital signs	General health problem	0	Problem	Active
		Weight = 85.5 kg	Vital signs	Obesity WHO grade II	100	Diagnosis	Active
		Height = 168 cm	Vital signs	General health problem	0	Problem	Active
		BMI = 30.5	Vital signs	Obesity WHO grade II	100	Diagnosis	Active
		S: Still experiencing urinary symptoms O: Urine culture: Escherichia coli resistant to current antibiotic A: Acute bacterial cystitis P: Switch antimicrobial therapy	Progess notes	Urinary tract infection	10	Diagnosis	Active
			Problem/Diagnosis	Acute bacterial cystitis	10	Diagnosis	Active
		Amoxicillin 500 mg / Clavulanate 125 mg film-coated tablet 3x1	Medication	Acute bacterial cystitis	10	Diagnosis	Active

Figure 2 – Chronological by contact: Entries grouped per contact and ordered by time

User Story 2 – Episode pivot

As a healthcare professional, I would like to be able to select an episode of care (health problem = diagnosis/problem) and then view all medical record entries relating to that health problem in the patient's complete CDR.



Date	Contact	Detail entry	Section	Health problem Episode of Care	ID	Condition	Clinical Status
15.11.2024	Consultation		Problem/Diagnosis	Dysuria	10	Problem	Active
15.11.2024	Consultation	Blood pressure 140/90	Vital signs	Dysuria	10	Problem	Active
15.11.2024	Consultation	Heart rate 75	Vital signs	Dysuria	10	Problem	Active
15.11.2024	Consultation	Weight = 86 kg	Vital signs	Dysuria	10	Problem	Active
15.11.2024	Consultation	Height = 168 cm	Vital signs	Dysuria	10	Problem	Active
15.11.2024	Consultation	BMI = 30.5	Vital signs	Dysuria	10	Problem	Active
15.11.2024	Consultation	Urinalysis – dipstick: Leukocyte esterase: 3+, Nitrite: positive, Blood: 1+, Protein: trace, pH: 6.0	Laboratory results	Dysuria	10	Problem	Active
15.11.2024	Consultation	S: Burning sensation during urination for 3 days. First episode. Denies visible blood in urine O:pyuria A: Suspected cystitis P: antibiogram, Started treatment with trimethoprim–sulfamethoxazole. Therapy adjusted per antibiogram	Progess notes	Dysuria	10	Problem	Active
15.11.2024	Consultation		Problem/Diagnosis	Urinary tract infection	10	Diagnosis	Active
15.11.2024	Consultation	Trimethoprim–sulfamethoxazole (TMP 160 mg / SMX 800 mg) PO 2x1	Medication	Urinary tract infection	10	Diagnosis	Active
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17.11.2024	Consultation	S: Still experiencing urinary symptoms O: Urine culture: Escherichia coli resistant to current antibiotic A: Acute bacterial cystitis P: Switch antimicrobial therapy	Progess notes	Urinary tract infection	10	Diagnosis	Active
17.11.2024	Consultation		Problem/Diagnosis	Acute bacterial cystitis	10	Diagnosis	Active
17.11.2024	Consultation	Amoxicillin 500 mg / Clavulanate 125 mg film-coated tablet 3x1	Medication	Acute bacterial cystitis	10	Diagnosis	Active
23.11.2024	Consultation	Urinalysis – dipstick: Leukocyte esterase: negative, Nitrite: negative, Blood: negative, Protein: negative, pH: 6.0	Laboratory results	Acute bacterial cystitis	10	Diagnosis	Active
23.11.2024	Consultation	S: Patient reports resolution of urinary symptom O: Urine culture: Consistent with sterilised urine. A: Acute bacterial cystitis – resolved P: Discontinue antimicrobial therapy (completed full course). Routine follow-up at next scheduled chronic-care visit; no additional testing required unless symptomatic.	Progess notes	S/P acute bacterial cystitis	10	Diagnosis	Resolved
25.11.2024	Telephone	S: Patient reports slight urge to urinate and wants to know if he needs	Progess notes	S/P acute bacterial cystitis	10	Diagnosis	Resolved

Figure 3 – Episode pivot: Entries linked to a selected episode across contacts

User Story 3 – Clinical section-based view (excerpt)

As a healthcare professional, I would like to see all medical record entries in a patient's CDR organised according to the standard clinical sections of a medical record, i.e. medical history, physical examination, diagnostics, therapy, risk factors, progress notes, etc.

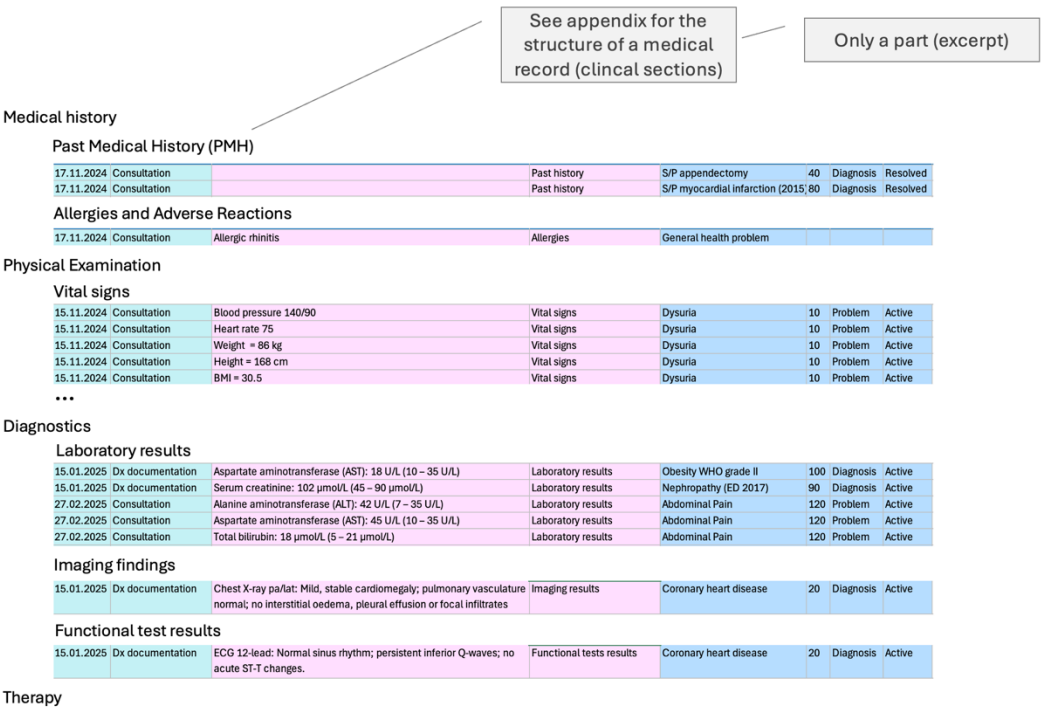


Figure 4 – Section-based view (excerpt): Entries organised by clinical sections

User Story 4 – Dual grouping

As a healthcare professional, I would like the grouping to be according to the standard clinical sections of a medical record, with the chronological grouping by contacts or with the corresponding record entries for the selected health problem displayed.

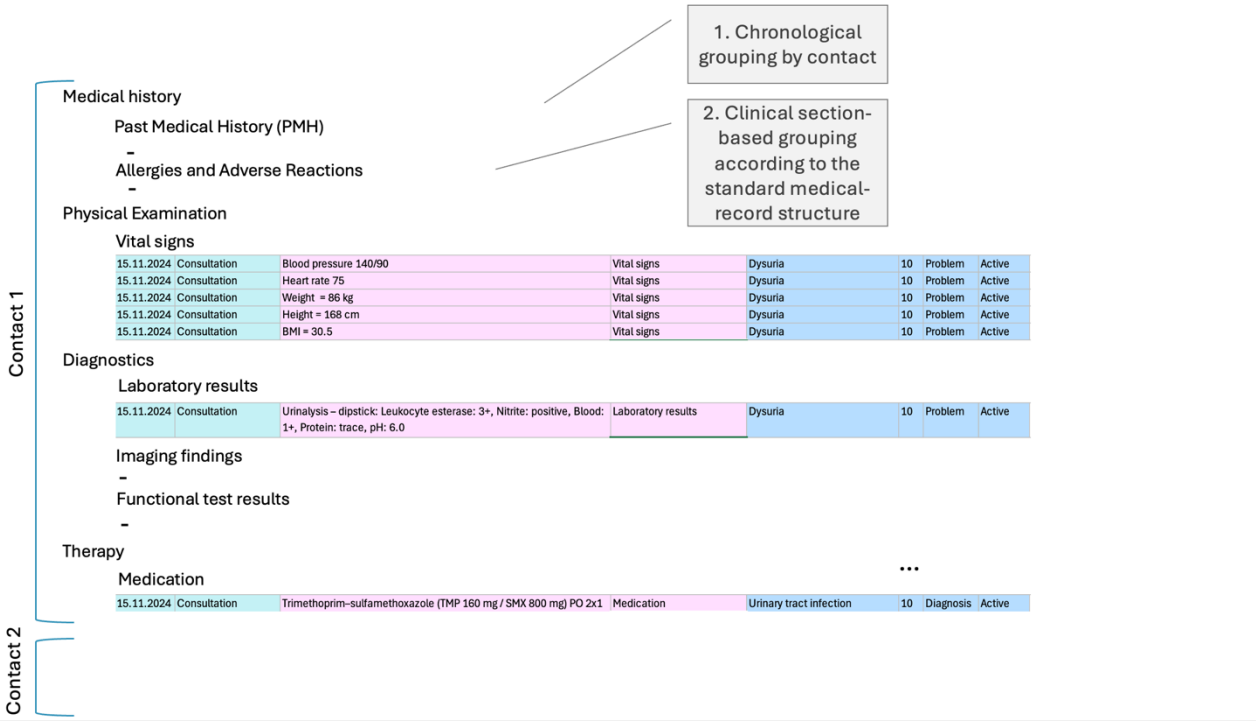


Figure 5 – Dual grouping: Chronological by contact and by clinical section

User Story 5 – Linear episode list

As a healthcare professional, I would like to have a linear list of all patient's recorded health problems (= diagnosis/problem = episode-of-care).

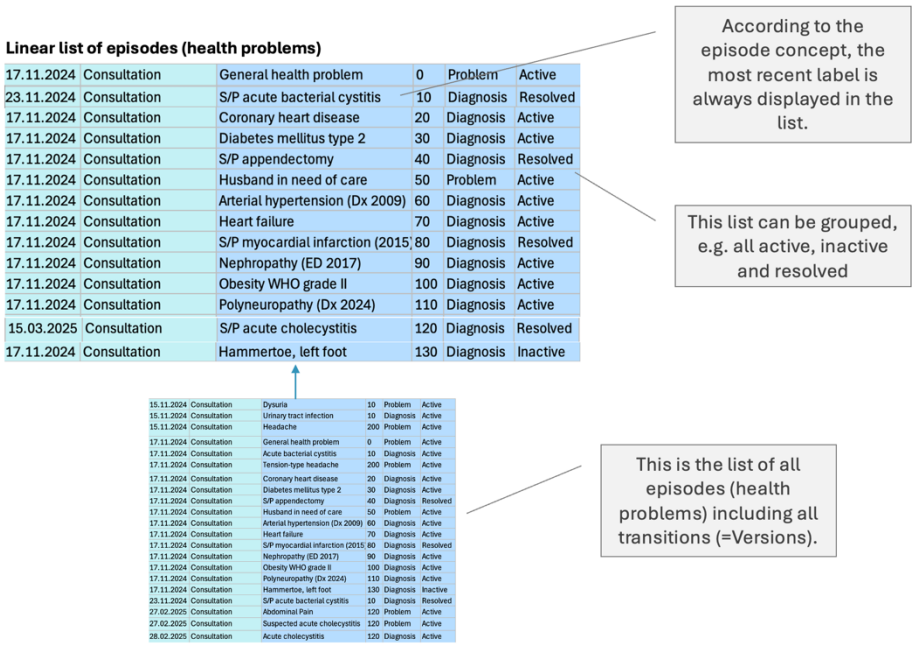


Figure 6 – Linear episode list: All episodes with key status/progression

User Story 6 – Master diagnosis & problem list

As a healthcare professional, I would like to be able to create a hierarchical 'diagnosis and problem list' for each patient, organised in a tree structure according to Weed's problem list, which I could then adapt continuously.

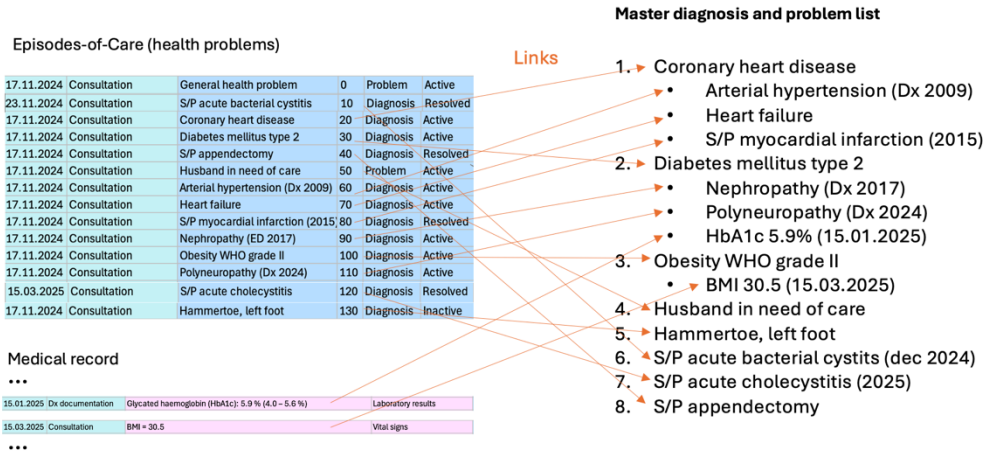


Figure 7 – Master diagnosis & problem list: Hierarchical (Weed-style) view

User Story 7 – SOAP progress notes

As a healthcare professional, I would like to be able to create progress notes according to the SOAP scheme for each health problem (= diagnosis/problem = episode-of-care) or for several health problems at the same time.

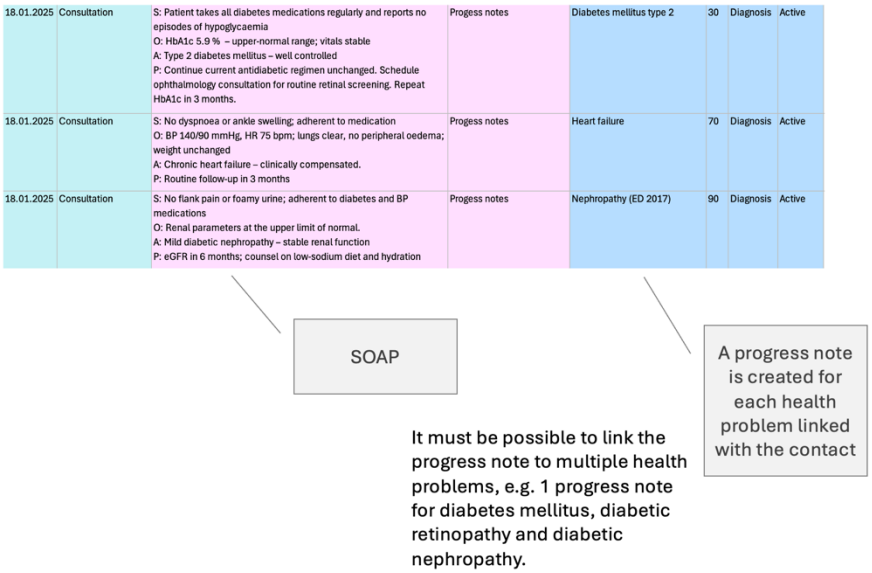


Figure 8 – SOAP progress notes: Note linked to one or multiple episodes

User Story 8 – Dashboard

As a healthcare professional, I would like a dashboard for each patient containing various widgets. Each widget would display medical record entries for a single section, in accordance with the standard clinical sections of a medical record.

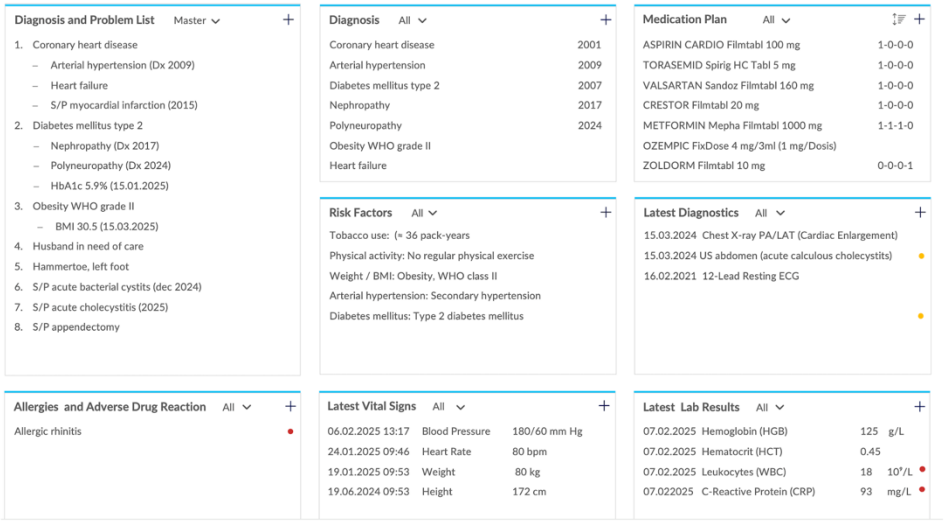


Figure 9 – Patient dashboard: Section-specific widgets as derived views